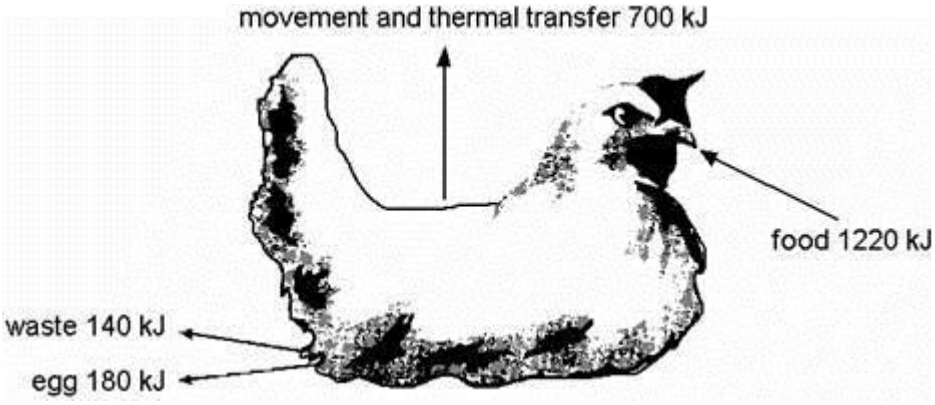
**Q1.**

The drawing shows what happens to most of the energy in the food that a hen eats in one day.




(a) In the cells of the hen's body, energy is released from food by respiration
Complete the word equation for this process.

Glucose + …… → …… + ……..

1 mark

(b) (i) Calculate the total energy which remains in the body of the hen

.....

..... kJ

1 mark

(ii) What is this energy used for?

.....

.....

1 mark

(c) Suggest how farmers might reduce the amount of energy which hens lose each day by thermal transfer.

.....

.....

1 mark

(d) Farmers can reduce the amount of energy which is transferred by movement and thermal transfer from hens. Suggest **two** reasons why this is cost-effective.

1. .....

.....

2. .....

.....

2 marks

Maximum 6 marks

Mark schemes

## Q1.

(a) (i) oxygen → carbon dioxide + water

***all three*** *substances are required for the mark*

*the products may be in either order*

**1 (L7)**

(b) (i) 200

**1 (L6)**

(ii) any **one** from

- for growth
- for producing meat
- for storage
- for making fat

*accept 'for active transport'* ***or*** *'for keeping internal organs running'* ***or*** *'for metabolism' do* ***not*** *accept 'for keeping the hen warm'* ***or*** *'for breathing'*

**1 (L7)**

(c) keep the hens warm

*accept 'keep them inside '****or*** *'keep them cooped up '****or*** *'stop them moving'* ***or*** *'keep them in deep litter'* ***or*** *'keep them all together'*

**1 (L7)**

(d) any **two** from

- more energy is available for producing eggs

  *accept 'get more eggs'* ***or*** *'get bigger eggs'*

- more energy is available for growth

  *accept 'get more meat'* ***or*** *'get bigger hens'*

- the hens do not have to be fed so much

  *accept 'they can use smaller buildings for the same number of hens'*

**2 (L7)**

**[6]**